Assignment 1

Data Types & Fixed Arrays

Part A: Data Types

1. **Declare the following variables:**

   a) A **4-bit signed packed logic vector** named a
   b) An **8-bit unsigned byte** named b
   c) A **16-bit shortint signed** named c
   d) A **bit [3:0]** vector named d
   e) A **real** variable named r
   f) A **string** called myString initialized to "SV Assignment1"

2. **Write an initial block** to:
   - Assign -3 to a and print its decimal value.
   - Assign 250 to b and print its value.
   - Assign -12345 to c and print it.
   - Assign 4'b1101 to d and print its binary after 5ns delay.
   - Assign 3.14159 to r and print it after 15ns delay.
   - Print the string myString.

Part B: Fixed Arrays: - in the same file

3. **Declare and initialize the following fixed arrays:**

   a) A **1D unpacked array of int type** with values 1,2,3,4,5
   b) A **packed array of logic type of three elements two bit each**. Assign all elements to **2'b10.**
   c) A **1D unpacked array has 4 elements of type byte** with values **10, 20, 30, 40**.
   d) Print all the arrays content in separate display statement.

5. **Declare a struct type** called student_t with:
   - string name
   - int id
   - byte grade

Then:

a) Declare a **fixed array of 3 students of type** student_t called students.
b) Assign the following:

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```
students[0] = {"Ali", 1, 85};
students[1] = {"Sara", 2, 90};
students[2] = {"Omar", 3, 78};
```

c) print each student's details in the format: `Name: <name>, ID: <id>, Grade: <grade>`

6. **Declare a union type** called data_u with:
   - int i;
   - byte b;

   Then:

   a) Declare a variable of type data_u called data
   b) Assign 32'h12345678 to data.i
   c) Print data.i and data.b as hexadecimal.
   d) Assign 8'h12 to data.i
   e) Print data.i and data.b as hexadecimal.

Deliverables:

U have to deliver a Zip Folder contains

- **ONE PDF** contains:
  - Snippets for the implemented code.
  - Snippet for the waveform shows the values for created variables
  - Snippets for the logs show the all printed values.

- The implemented **.sv** file.

The delivered zip file must be named like **your_name_Assignment_1.rar** for example:
Hassan_Khaled_Assignment_1.rar

Deadline:

The deadline is Thu 10/7 at 4 PM

Good Luck

Hassan Khaled